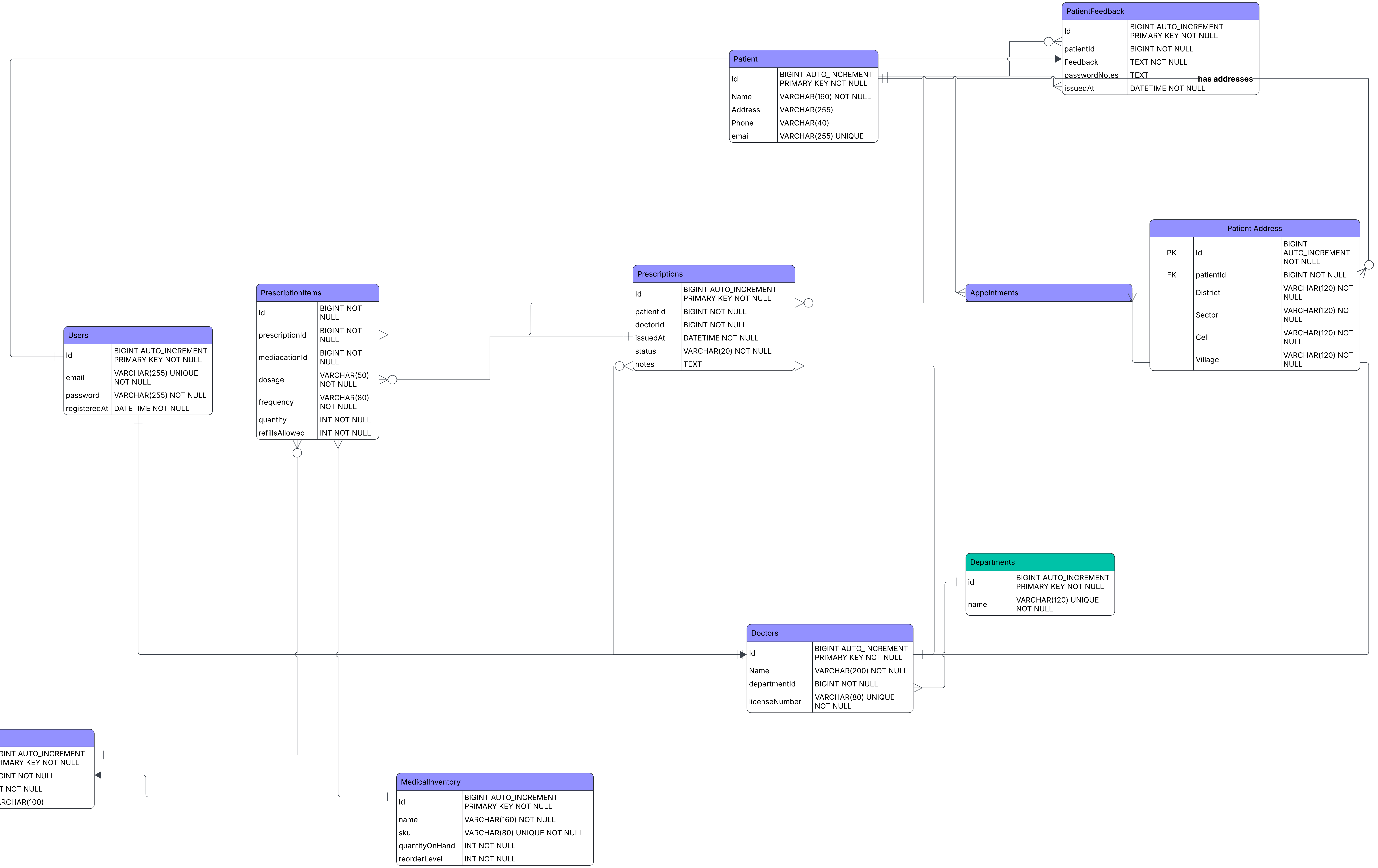
IndexDefinitions	
SQL_1	ALTER TABLE `Doctors` ADD CONSTRAINT `fk_doctors_department` FOREIGN KEY (`departmentId`) REFERENCES `Departments` (`id`);
SQL_2	ALTER TABLE `Patient Address` ADD CONSTRAINT `fk_patientaddress_patient` FOREIGN KEY (`patientId`) REFERENCES `Patient` (`Id`);
SQL_3	ALTER TABLE `Appointments` ADD CONSTRAINT `fk_appointments_patient` FOREIGN KEY (`patientId`) REFERENCES `Patient` (`Id`), ADD CONSTRAINT `fk_appointments_doctor` FOREIGN KEY (`doctorId`) REFERENCES `Doctors` (`Id`);
SQL_4	ALTER TABLE `Prescriptions` ADD CONSTRAINT `fk_prescriptions_patient` FOREIGN KEY (`patientId`) REFERENCES `Patient` (`Id`), ADD CONSTRAINT `fk_prescriptions_doctor` FOREIGN KEY (`doctorId`) REFERENCES `Doctors` (`Id`);
SQL_5	ALTER TABLE `PatientFeedback` ADD CONSTRAINT `fk_patientfeedback_patient` FOREIGN KEY (`patientId`) REFERENCES `Patient` (`Id`);
SQL_6	ALTER TABLE `MedicalItems` ADD CONSTRAINT `fk_medicalitems_inventory` FOREIGN KEY (`inventoryId`) REFERENCES `MedicalInventory` (`Id`);
SQL_7	ALTER TABLE `PrescriptionItems` ADD CONSTRAINT `fk_prescriptionitems_prescription` FOREIGN KEY (`prescriptionId`) REFERENCES `Prescriptions` (`Id`), ADD CONSTRAINT `fk_prescriptionitems_medication` FOREIGN KEY (`medicationId`) REFERENCES `MedicalInventory` (`Id`);
SQL_8	CREATE INDEX `idx_patient_name` ON `Patient` (`Name`);
SQL_9	CREATE INDEX `idx_doctors_department` ON `Doctors` (`departmentId`);
SQL_10	CREATE INDEX `idx_appointments_date` ON `Appointments` (`appointmentDate`);
SQL_11	CREATE INDEX `idx_appointments_patient` ON `Appointments` (`patientId`);
SQL_12	CREATE INDEX `idx_appointments_doctor` ON `Appointments` (`doctorId`);
SQL_13	CREATE INDEX `idx_prescriptions_patient` ON `Prescriptions` (`patientId`);
SQL_14	TEXT

Relationship notation

- 1:1 | exactly one ↔ exactly one
- 1:M | one ↔ many
- 0..1 | zero or one
- 0..M | zero or many
- M:N | many ↔ many
- Crow's foot | many side
- Circle | optional / zero
- Bar | exactly one



Patient	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
Name	VARCHAR(160) NOT NULL
Address	VARCHAR(255)
Phone	VARCHAR(40)
email	VARCHAR(255) UNIQUE

PatientFeedback	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
patientId	BIGINT NOT NULL
Feedback	TEXT NOT NULL
passwordNotes	TEXT
issuedAt	DATETIME NOT NULL

Patient Address		
PK	Id	BIGINT AUTO_INCREMENT NOT NULL
FK	patientId	BIGINT NOT NULL
	District	VARCHAR(120) NOT NULL
	Sector	VARCHAR(120) NOT NULL
	Cell	VARCHAR(120) NOT NULL
	Village	VARCHAR(120) NOT NULL

Appointments

Users	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
email	VARCHAR(255) UNIQUE NOT NULL
password	VARCHAR(255) NOT NULL
registeredAt	DATETIME NOT NULL

PrescriptionItems	
Id	BIGINT NOT NULL
prescriptionId	BIGINT NOT NULL
mediacationId	BIGINT NOT NULL
dosage	VARCHAR(50) NOT NULL
frequency	VARCHAR(80) NOT NULL
quantity	INT NOT NULL
refillsAllowed	INT NOT NULL

Prescriptions	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
patientId	BIGINT NOT NULL
doctorId	BIGINT NOT NULL
issuedAt	DATETIME NOT NULL
status	VARCHAR(20) NOT NULL
notes	TEXT

Departments	
id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
name	VARCHAR(120) UNIQUE NOT NULL

Doctors	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
Name	VARCHAR(200) NOT NULL
departmentId	BIGINT NOT NULL
licenseNumber	VARCHAR(80) UNIQUE NOT NULL

MedicalItems	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
inventoryId	BIGINT NOT NULL
quantity	INT NOT NULL
category	VARCHAR(100)

MedicalInventory	
Id	BIGINT AUTO_INCREMENT PRIMARY KEY NOT NULL
name	VARCHAR(160) NOT NULL
sku	VARCHAR(80) UNIQUE NOT NULL
quantityOnHand	INT NOT NULL
reorderLevel	INT NOT NULL